

THE UNIVERSITY OF HONG KONG
DEPARTMENT OF MATHEMATICS

ARIN7013: Numerical methods in artificial intelligence

May 19, 2025

6:30 p.m. – 9:00 p.m.

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

Answer ALL 5 questions

Note: You should always give precise and adequate explanations to support your conclusions. Clarity of presentation of your argument counts. So **think carefully before you write.**

1. (20 points) Define the order n tridiagonal matrix $\mathbb{A}_n \in \mathbb{R}^{n \times n}$,

$$\mathbb{A}_n := \begin{bmatrix} 2 & -1 & & & \\ -1 & 2 & -1 & & \\ & -1 & 2 & \ddots & \\ & & \ddots & \ddots & -1 \\ & & & -1 & 2 \end{bmatrix}.$$

- (a) Find a general formula for the LU factorization $\mathbb{A}_n = \mathbb{L}\mathbb{U}$. Hint: Consider the case $n = 3, 4, 5$ and then guess the general pattern and verify it.
- (b) Let $n = 5$, and $\mathbf{b} := [3; -2; 2; -2; 1]$. Solve $\mathbb{A}_5 \mathbf{x} = \mathbf{b}$.
- (c) Let $n = 5$, and $\mathbf{b} := [3; -2; 2; -2; 1]$. Solve $\mathbb{A}_5 \mathbf{x} = \mathbf{b}$ using Gauss-Jacobi method with $\mathbf{x}_0 = [0; 0; 0; 0; 0]$. Present the solutions for the first three iterations.
- (d) For any n , and any $\mathbf{b} \in \mathbb{R}^n$, will Gauss-Jacobi method converge? If so, can you explain it.
2. (25 points) Answer the following questions.

- (a) (Cross-correlation) Let

$$I := \begin{bmatrix} 0 & 2 & 1 \\ 4 & 3 & 5 \\ 8 & 7 & 6 \end{bmatrix}, K := \begin{bmatrix} 0 & 1 \\ 2 & 3 \end{bmatrix}.$$

What is $I * K$?

- (b) (Padding) Let

$$I := \begin{bmatrix} 2 & 1 & 0 \\ 4 & 3 & 5 \\ 6 & 8 & 7 \end{bmatrix}, K := \begin{bmatrix} 0 & 1 \\ 2 & 3 \end{bmatrix}.$$

Padding I with zeros to make a matrix $\tilde{I}$ of size 5×5 such that its first row, last row, first column and last column are all zeros. What is $\tilde{I} * K$?

- (c) (Positive definite kernel) Let $\chi := \mathbb{R}^d$ and $\gamma > 0$. Prove that the function $K : \chi^2 \rightarrow \mathbb{R}$ defined by:

$$\forall (\mathbf{x}, \mathbf{x}') \in \chi^2 : K(\mathbf{x}, \mathbf{x}') = e^{-\gamma \|\mathbf{x} - \mathbf{x}'\|_2^2}$$

is positive definite.

(d) What is the complexity of calculating $\mathbb{A}\mathbb{B}$ for two $n \times n$ matrices $\mathbb{A}$ and $\mathbb{B}$.

(e) What is the complexity of calculating $\mathbb{A}\mathbb{B}$ where $\mathbb{A}$ is $m \times n$ and $\mathbb{B}$ is $n \times r$.

3. (30 points) Given a data matrix $\mathbb{X} = [\mathbf{x}_1 \ \mathbf{x}_2 \ \mathbf{x}_3 \ \mathbf{x}_4] \in \mathbb{R}^{6 \times 4}$, where

$$\mathbf{x}_1 = \begin{bmatrix} 2 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{x}_2 = \begin{bmatrix} 0 \\ -1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{x}_3 = \begin{bmatrix} 0 \\ \frac{1}{\sqrt{2}} \\ 0 \\ 0 \\ \frac{1}{\sqrt{2}} \\ 0 \end{bmatrix}, \quad \mathbf{x}_4 = \begin{bmatrix} 0 \\ 0 \\ \frac{1}{\sqrt{2}} \\ 0 \\ 0 \\ \frac{1}{\sqrt{2}} \end{bmatrix}.$$

(a) Find the first principal component $\mathbf{w} \in \mathbb{R}^6$, and then compute the mean square error,

$$\frac{1}{4} \sum_{i=1}^4 \|\mathbf{x}_i - \langle \mathbf{x}_i, \mathbf{w} \rangle \mathbf{w}\|_2^2.$$

Here, you do not need to center the samples before applying PCA for the sake of simplicity. $\langle \mathbf{x}_i, \mathbf{w} \rangle := \mathbf{x}_i^\top \mathbf{w}$.

(b) Let $\mathbf{v}^{(0)} = (\frac{1}{\sqrt{2}}, 0, 0, \frac{1}{\sqrt{2}})^T$ be the initial eigenvector. Find the largest eigenvalue of the matrix $\mathbb{X}^T \mathbb{X}$ starting with $\mathbf{v}^{(0)}$ using the power iteration. Display the results for the first three iterations.

(c) Write down the SVD decomposition for $\mathbb{X}$.

(d) Let $K(\cdot, \cdot) : \mathbb{R}^6 \times \mathbb{R}^6 \rightarrow \mathbb{R}$ be given by

$$K(\mathbf{x}, \mathbf{x}') = |\langle \mathbf{x}, \mathbf{x}' \rangle|^2.$$

Note that $K(\cdot, \cdot)$ is a positive kernel, and thus there is a corresponding Hilbert space, denoted by $(\mathcal{H}, \langle \cdot, \cdot \rangle_{\mathcal{H}})$.

i. Use this kernel to perform kernel PCA, and find the first principal component $v \in \mathcal{H}$. Here, you do not need to center the samples before applying kernel PCA for the sake of simplicity.

ii. Recall that this first principal component v can be expressed as

$$v(\mathbf{x}) = \sum_{i=1}^4 a_i K(\mathbf{x}_i, \mathbf{x}),$$

satisfying $\langle v, v \rangle_{\mathcal{H}} = 1$. Calculate the mean square error,

$$\frac{1}{4} \sum_{i=1}^4 \|K(\mathbf{x}_i, \cdot) - \langle v, K(\mathbf{x}_i, \cdot) \rangle_{\mathcal{H}} v\|_{\mathcal{H}}^2.$$

- (e) Question 3(a) and 3(d)ii give two different approaches to find the first principal component of the same data. Can you compare the performance of these two approaches?
4. (10 points) Let $f(t, x) : (0, 1] \times [x_0 - b, x_0 + b]$ be a bivariate function with bounded partial derivatives up to order 3. Consider the following initial value problem,

$$\frac{dx}{dt} = f(t, x) \text{ for } t \in (0, 1], \quad x(0) = x_0. \quad (1)$$

For $N \in \mathbb{N}$, we divide the interval $[0, 1]$ by the $N + 1$ mesh points,

$$0 = t_0, t_1 = h, \dots, t_N = Nh = 1,$$

with $h := 1/N$ as the step size.

- (a) Write down the θ -method with $\theta = 0.5$ for (1). Is it an explicit method?
- (b) Repeat the global error estimate for the explicit Euler method to derive the global error estimate for the θ -method with $\theta = 0.5$.
5. (15 points) Consider a convolutional neural network that is used to classify images into two classes. The structure of the network is as follows:
- INPUT. 100×100 grayscale images: X
 - LAYER 1. Convolutional layer with one 5×5 convolutional filter K . Here, we set the stride length=1. In order to capture the boundary of the input image, we use padding by adding 2 rows and 2 columns with zero elements around it, which is denoted by $\tilde{X}$.

- LAYER 2. A max pooling layer of size 2×2 with stride length=2 to reduce the dimension.
 - LAYER 3. Flatten layer.
 - LAYER 4. Fully connected Layer with 100 nodes (with Relu activation function).
 - LAYER 5. Single output unit: y
- (a) What is the output shape from each layer, i.e., what is the dimension of data from each layer?
- (b) How many weights does this network have? You can ignore the bias in this question.
- (c) Is there any way to modify this convolutional neural network as a residual network? Please explain your approach, and explain possible advantage of this modification compared with the plain convolutional neural network.

***** END OF PAPER *****